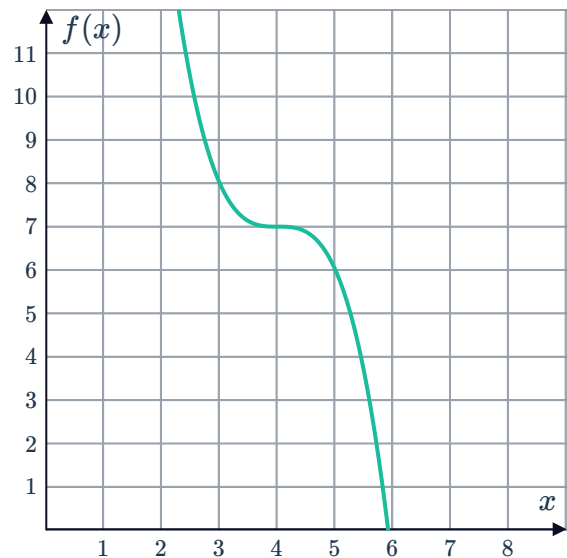


## Functions and gradient functions



- 1 Consider the graph of the function  $f(x) = -(x - 4)^3 + 7$ :

- a State the  $x$ -value of the stationary point of  $f(x)$ .
- b State the domain where  $f(x)$  is decreasing.



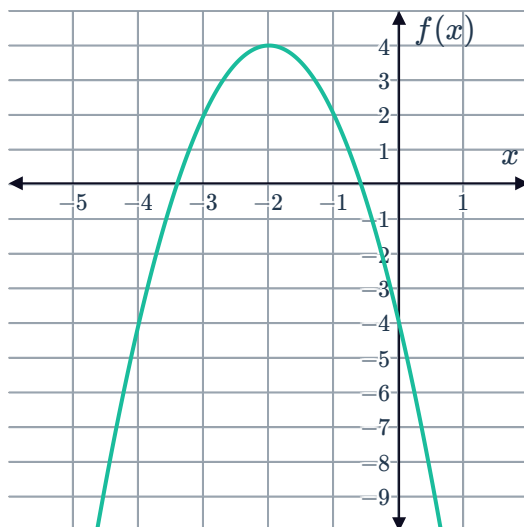
- 2 State whether the rate of change of the following functions is positive, negative, or zero for all values of  $x$ :

- a  $f(x) = -8$
- b  $f(x) = 4x - 1$
- c  $f(x) = -x + 10$

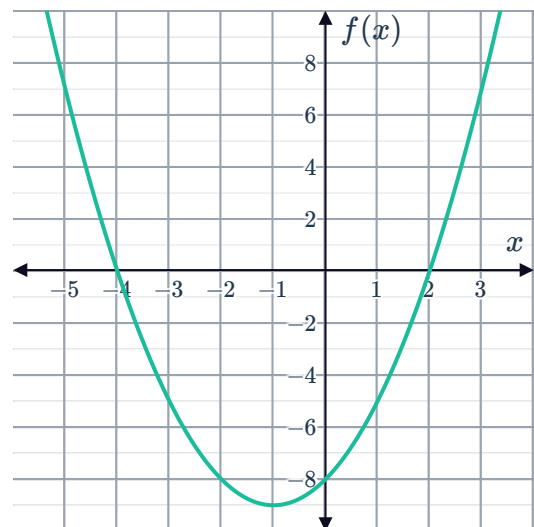
- 3 For each of the following functions:

- i State the  $x$ -value of the stationary point of  $f(x)$ .
- ii State the domain where  $f(x)$  is increasing.
- iii State the domain where  $f(x)$  is decreasing.

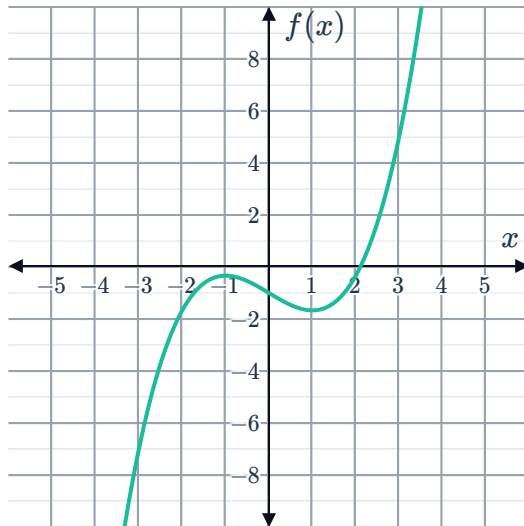
a  $f(x) = -2(x + 2)^2 + 4$



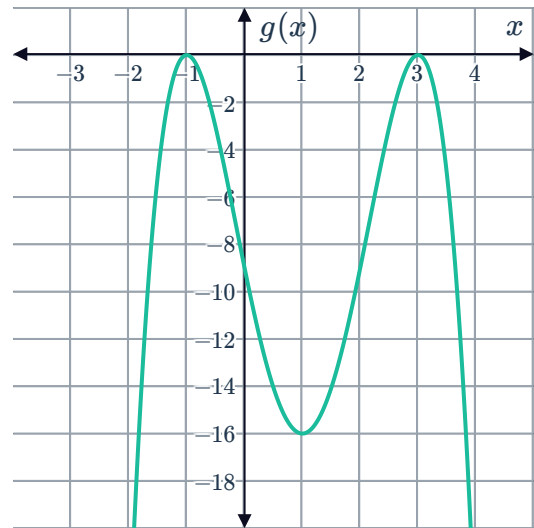
b  $f(x) = (x + 4)(x - 2)$



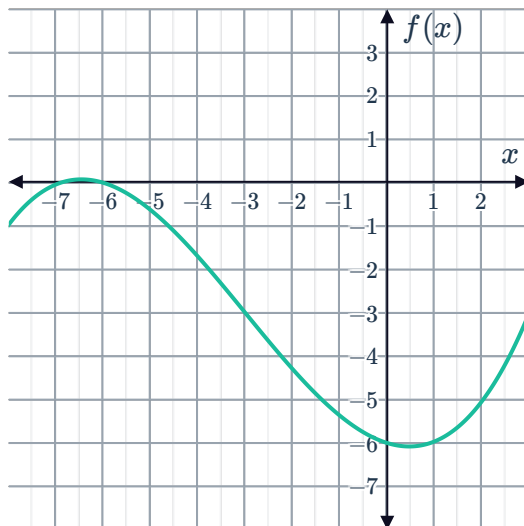
c  $f(x) = \frac{x^3}{3} - x - 1$



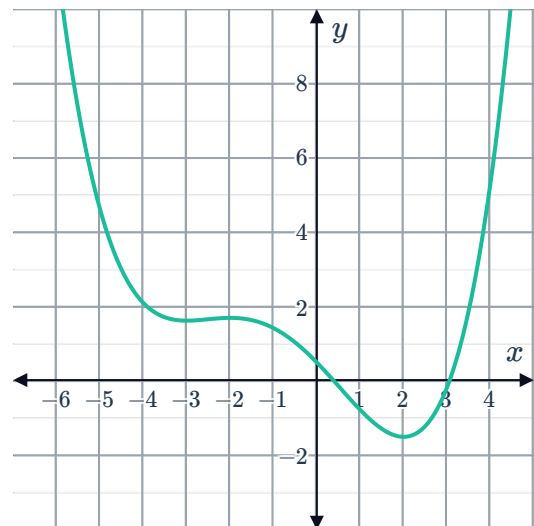
d  $g(x) = -(x-3)^2(x+1)^2$



e  $f(x)$



f  $f(x)$

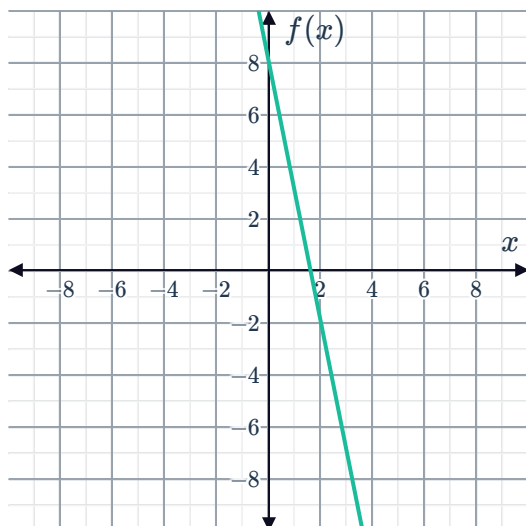


4 Consider the function  $y = 4x - 3$ .

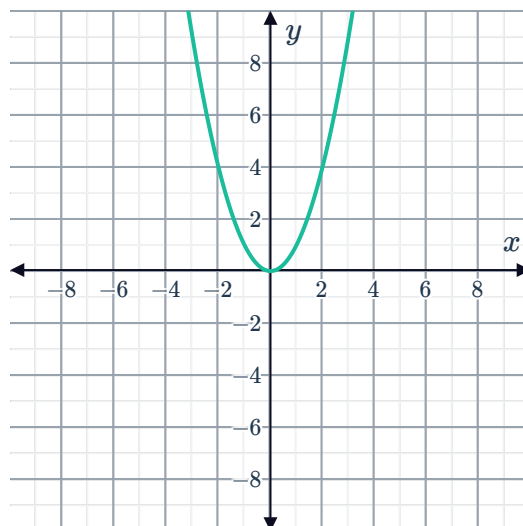
- Find the gradient function.
- Sketch the graph of the gradient function.

5 For each of the following functions, sketch  gradient function:

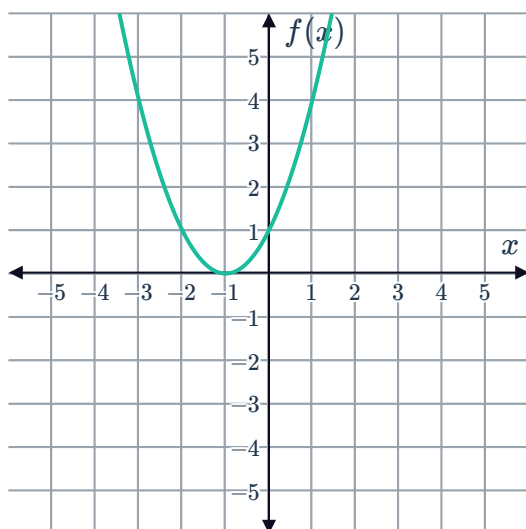
a  $f(x) = -5x + 8$



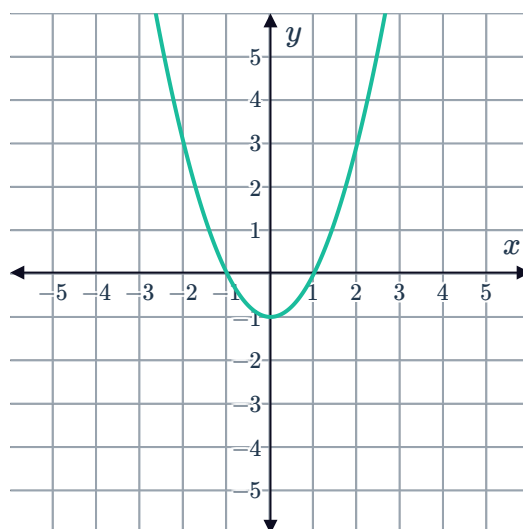
b  $y = x^2$



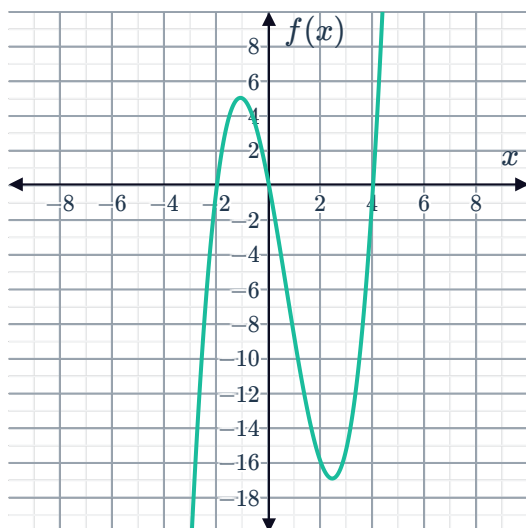
c  $f(x) = (x + 1)^2$



d  $y = x^2 - 1$



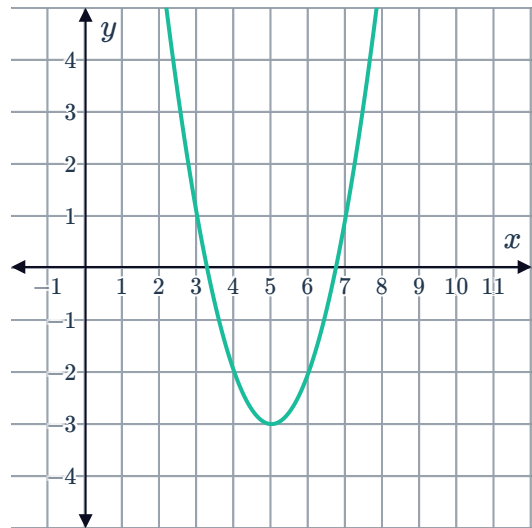
e  $f(x) = x(x - 4)(x + 2)$



6 Consider the function  $y = (x - 5)^2 - 3$  graphed below:

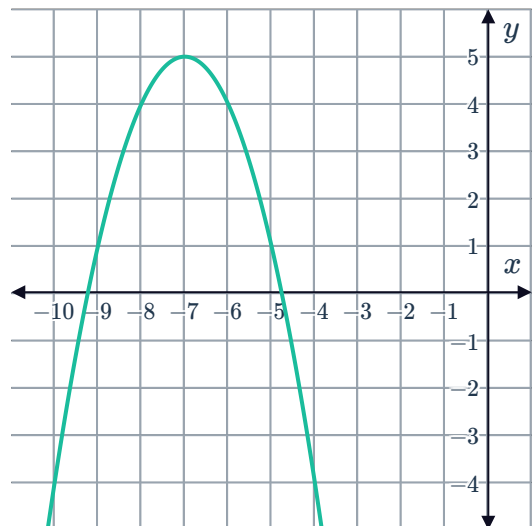


- a State the  $x$ -intercept of the gradient function.
- b For  $x < 5$ , state whether the values of the gradient function are above or below the  $x$ -axis.
- c For  $x > 5$ , state whether the values of the gradient function are above or below the  $x$ -axis.



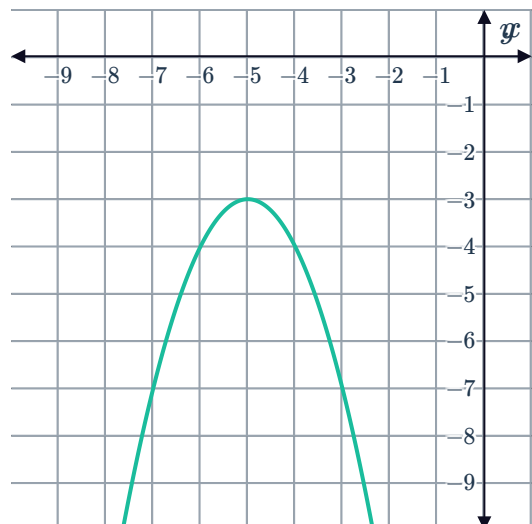
7 Consider the function  $y = -(x + 7)^2 + 5$  graphed below:

- a State the  $x$ -intercept of the gradient function.
- b For  $x < -7$ , state whether the values of the gradient function are above or below the  $x$ -axis.
- c For  $x > -7$ , state whether the values of the gradient function are above or below the  $x$ -axis.



8 Consider the function  $y = -x^2 - 10x - 28$  graphed below:

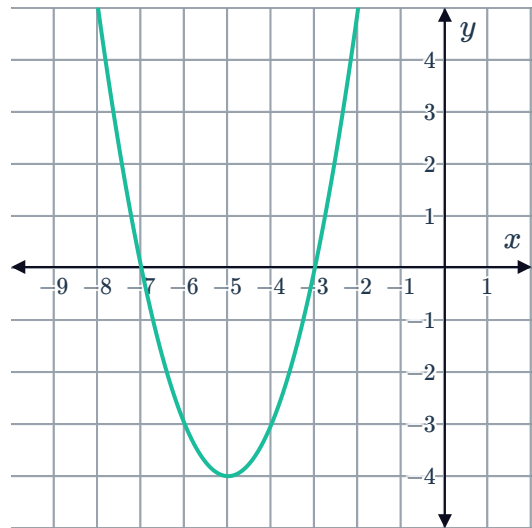
- a State the  $x$ -intercept of the gradient function.
- b For  $x < -5$ , state whether the values of the gradient function are above or below the  $x$ -axis.
- c For  $x > -5$ , state whether the values of the gradient function are above or below the  $x$ -axis.



9 Consider the function  $y = x^2 + 10x + 21$  graphed below:

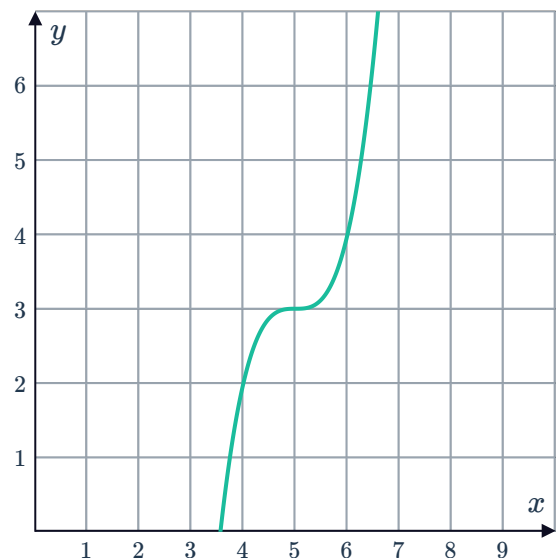


- a State the  $x$ -intercept of the gradient function.
- b For  $x < -5$ , state whether the values of the gradient function are above or below the  $x$ -axis.
- c For  $x > -5$ , state whether the values of the gradient function are above or below the  $x$ -axis.



10 Consider the function  $y = (x - 5)^3 + 3$ :

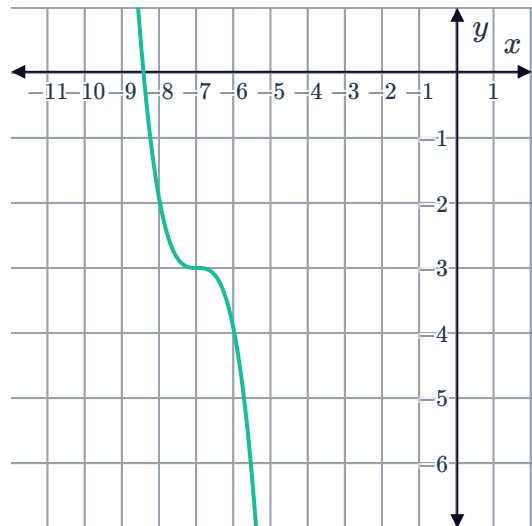
- a State the coordinates of the point of inflection.
- b State the gradient of the curve at this point.
- c What feature does the gradient function have at  $x = 5$ ?
- d For  $x < 5$ , state whether the values of the gradient function are above or below the  $x$ -axis.
- e For  $x > 5$ , state whether the values of the gradient function are above or below the  $x$ -axis.
- f What type of point is on the gradient function at  $x = 5$ ?



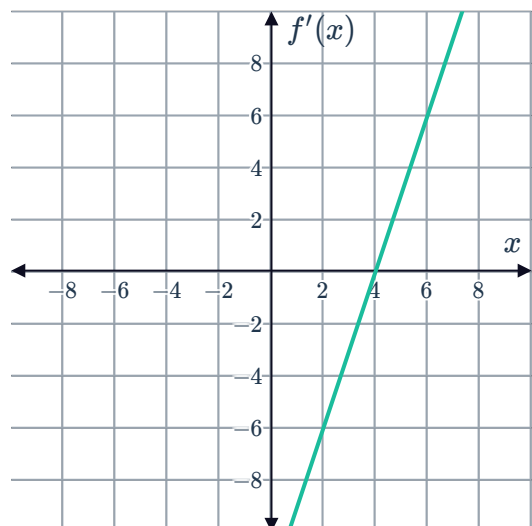
11 Consider the function  $y = -(x + 7)^3 - 3$ :



- a State the coordinates of the point of inflection.
- b State the gradient of the curve at this point.
- c What feature does the gradient function have at  $x = -7$ ?
- d For  $x < -7$ , state whether the values of the gradient function are above or below the  $x$ -axis.
- e For  $x > -7$ , state whether the values of the gradient function are above or below the  $x$ -axis.
- f What type of point is on the gradient function at  $x = -7$ ?



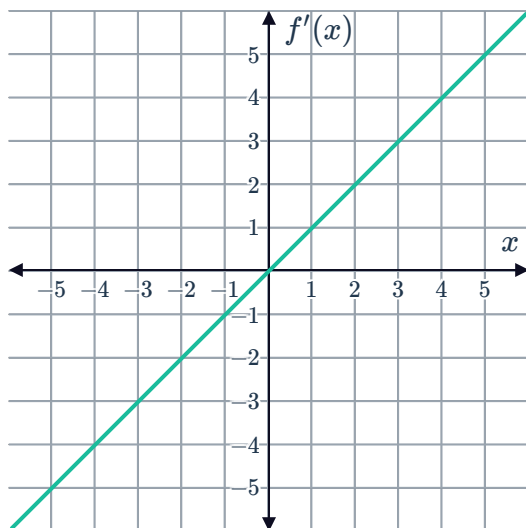
12 Consider the graph of the gradient function  $f'(x)$ :  
What can be said about the graph of  $f(x)$ ?



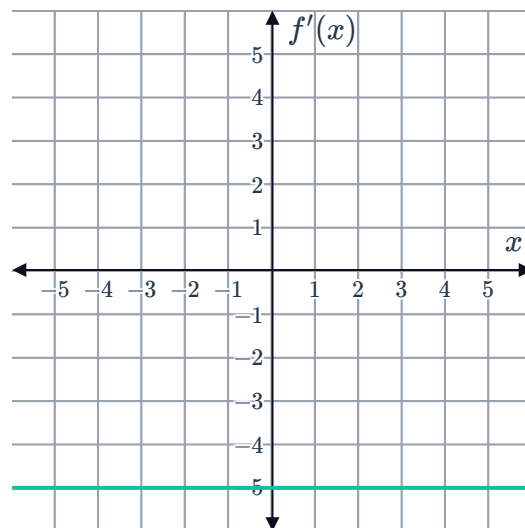
13 For each of the following gradient functions, sketch a graph of a possible original function:



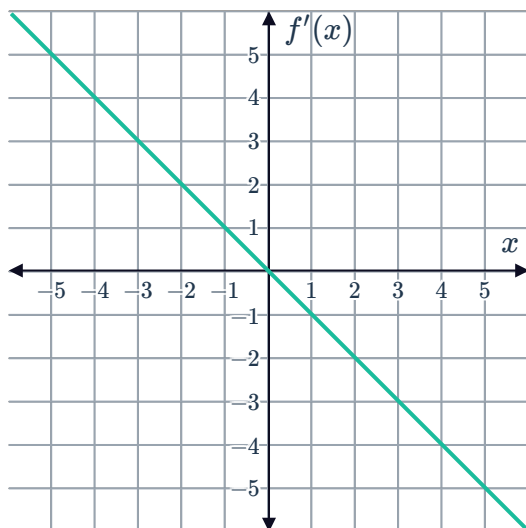
a



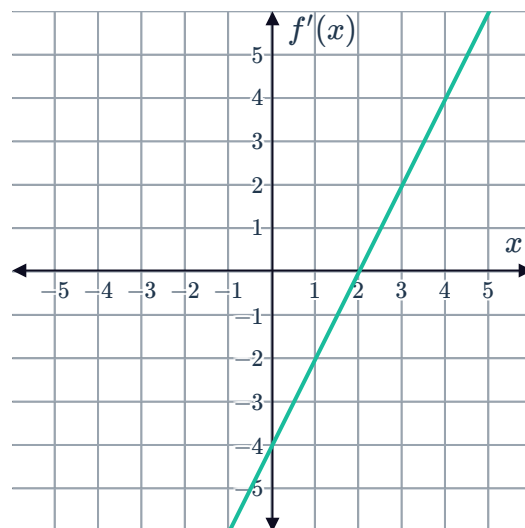
b



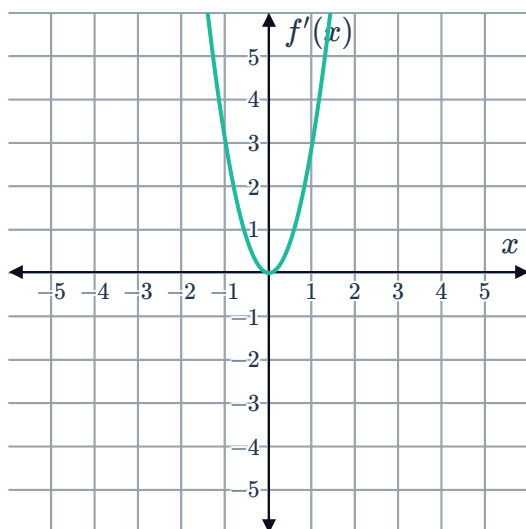
c



d



e





## Types of stationary points

**14** State the type of point that matches the following descriptions:

- a** A point where the curve changes from decreasing to increasing.
- b** A point where the curve changes from increasing to decreasing.
- c** A point where the tangent is horizontal and the concavity changes about the point.

**15** For each of the following functions:

- i** Find the derivative.
- ii** Find the coordinates of any stationary points.
- iii** Classify each stationary point.

**a**  $y = -6x^2 + 84x - 29$

**b**  $y = x^3 - 21x^2 + 144x - 19$

**c**  $f(x) = (x + 3)^2(x + 6)$

**d**  $f(x) = (x + 5)^3 + 4$

**e**  $f(x) = -\frac{x^3}{3} + \frac{13x^2}{2} - 30x + 10$

**f**  $f(x) = (4x + 5)(x + 1)$

**g**  $f(x) = 134 - 300x + 240x^2 - 64x^3$

**h**  $f(x) = (x^2 - 9)^2 + 4$

**16** Consider the parabola with equation  $y = 5 + x - x^2$ .

- a** Find the coordinates of the vertex of the parabola.
- b** State the gradient of the tangent to the parabola at the vertex.
- c** What type of stationary point is at the vertex of this parabola?

**17** Consider the function  $f(x) = x^2 + 4x + 9$ .

- a** Find an equation for the gradient function  $f'(x)$ .
- b** State the interval in which the function is increasing.
- c** State the interval in which the function is decreasing.
- d** Find the coordinates of the stationary point.
- e** Classify the stationary point.



18 Consider the function  $f(x) = 4x^3 + 5x^2 -$  

- Find an equation for the gradient function  $f'(x)$ .
- Find the coordinates of the stationary points.
- Complete the table of values:

|         |      |                |                |     |     |
|---------|------|----------------|----------------|-----|-----|
| $x$     | $-2$ | $-\frac{5}{6}$ | $-\frac{1}{2}$ | $0$ | $1$ |
| $f'(x)$ |      | $0$            |                | $0$ |     |

- Hence determine the:
  - Local minimum
  - Local maximum
- Is  $-4$  the absolute minimum value of the function? Explain your answer.

19 Consider the function  $f(x) = 3x^2 - 54x + 241$ .

- Find  $f'(x)$ .
- Find the  $x$ -coordinate of the stationary point.
- Classify the stationary point.

## Graphs of functions

20 Sketch the linear function for which  $f(0) = 1$  and  $f'(2) = 3$ .

21 Sketch the quadratic function,  $f(x)$ , that satisfies the following conditions:

- |                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>a</b></p> <ul style="list-style-type: none"> <li><math>f(0) = -18</math></li> <li><math>f(3) = 0</math></li> <li><math>f(6) = 6</math></li> <li><math>f'(6) = 0</math></li> <li><math>f'(x) &gt; 0</math> for <math>x &lt; 6</math></li> </ul> | <p><b>b</b></p> <ul style="list-style-type: none"> <li><math>f(0) = 16</math></li> <li><math>f(2) = 0</math></li> <li><math>f(5) = -9</math></li> <li><math>f'(5) = 0</math></li> <li><math>f'(x) &lt; 0</math> for <math>x &lt; 5</math></li> </ul>     |
| <p><b>c</b></p> <ul style="list-style-type: none"> <li><math>f(0) = 5</math></li> <li><math>f(-2) = 0</math></li> <li><math>f'(3) = 0</math></li> <li><math>f'(x) &gt; 0</math> for <math>x &lt; 3</math></li> </ul>                                 | <p><b>d</b></p> <ul style="list-style-type: none"> <li><math>f(0) = 10</math></li> <li><math>f(-2) = 0</math></li> <li><math>f(-6) = -8</math></li> <li><math>f'(-6) = 0</math></li> <li><math>f'(x) &lt; 0</math> for <math>x &lt; -6</math></li> </ul> |

**22** Sketch the cubic function,  $f(x)$ , that satisfies the following conditions:



**a**

- $f(0) = 7$
- $f(-2) = 0$
- $f(-4) = -1$
- $f'(-4) = 0$
- $f'(x) > 0$  for  $x < -4$
- $f'(x) > 0$  for  $x > -4$

**b**

- $f'(2) = 0$
- $f'(-3) = 0$
- $f'(x) < 0$  for  $-3 < x < 2$
- $f'(x) > 0$  elsewhere

**23** Sketch the quartic function,  $f(x)$ , that satisfies the following conditions:

**a**

- $f'(-1) = 0$
- $f'(4) = 0$
- $f'(x) > 0$  for  $x > 4$
- $f'(x) < 0$  elsewhere

**b**

- $f(0) = 0$
- $f'(0) = 0$
- $f'(2) = 0$
- $f'(-2) = 0$
- $f'(x) > 0$  for  $x < -2, 0 < x < 2$
- $f'(x) < 0$  elsewhere

**24** Consider the equation of the parabola  $y = 3x^2 - 18x + 24$ .

**a** Find the  $x$ -intercepts.

**b** Find the  $y$ -intercept.

**c** Find  $\frac{dy}{dx}$ .

**d** Find the stationary point.

**e** Classify the stationary point.

**f** Sketch the graph of the parabola.

**25** For each of the following functions:

**i** Find the  $y$ -intercept.

**ii** Find the  $x$ -intercepts.

**iii** Find  $f'(x)$ .

**iv** Hence find the  $x$ -coordinates of the stationary points.

**v** Classify the stationary points.

**vi** Sketch the graph of the function.

**a**  $f(x) = 9x^2 + 18x - 16$

**b**  $f(x) = (4x + 5)^2(x - 1)$

**c**  $f(x) = (2x - 1)^2(1 - x)$

**d**  $f(x) = (x + 2)^3 - 1$

**e**  $f(x) = x^3 + 11x^2 + 24x$

**f**  $f(x) = (x^2 - 4)^2 + 4$

**g**  $f(x) = 2(x - 1)^3 - 16$

**h**  $f(x) = x^4 - 4x^3$

**i**  $f(x) = x^3 + 5x^2$

**j**  $f(x) = x^2(x - 3)^2$

**26** Sketch the graph of the following function showing all stationary points:



**a**  $y = 2x^3 - 12x^2 + 18x - 8$

**b**  $y = 3x^5 - 20x^3 + 6$

**c**  $y = 2x^3 + 3x^2 - 36x + 5$

**d**  $y = 20 + 4x^3 - x^4$

**e**  $y = -3x^4 + 16x^3 - 24x^2 + 30$

**f**  $y = x^3 - 6x^2 - 15x + 7$